ANN-based Chiller FDD Model for Considering Various Operating Conditions

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Abstract

Fault detection and diagnosis (FDD) for a chiller is important to prevent energy waste due to malfunction and to reduce maintenance costs. In recent years, with the development of machine learning, data driven FDD became popular due to the reduced time and expert knowledge required to develop the model. The chiller operates under various operating conditions. However, it is difficult to obtain eligible data under all operating conditions. For practical use, a model trained under limited operating conditions should have acceptable performance under untrained operating conditions. In this study, a chiller FDD model based on artificial neural network for considering various operating conditions was proposed. The proposed model generalizes well under various operating conditions as it can find underlying patterns in the data. Experiments with the data from ASHRAE 1043-RP show that the proposed model has improved generalization performance for operating conditions compared to the SVM-based model.

Keywords: Fault detection and diagnosis; Centrifugal chiller; Artificial neural network;

1. Introduction

The energy consumed by the HVAC system accounts for 50-60% of the total energy of the building [1]. Among them, chiller components account for more than 40% of the total energy used in buildings [2]. Also, it has a major impact on the occupants comfort conditions and maintenance costs. Various faults can increase energy usage of chiller [3].

Fault detection and diagnosis (FDD) is a technology that detects and responds to faults in the early stages. Applying the FDD techniques to chiller can decrease energy waste and enhance occupants comfort.

Recently, air conditioning systems are becoming larger and more complex due to the enlargement, complexity, and automation of buildings. However, the increase in size and complexity of related equipment makes it difficult for the operator to have a comprehensive understanding of the system, and when faults occur, it requires expert knowledge and skilled experience, making maintenance difficult.

The data driven FDD methods started from the end of the 1990s and became more and more popular in the last decade due to the complexity of HVAC system. The chiller operates under various operating and load conditions. Obtaining normal and fault data under all operating conditions for training the FDD model requires a lot of time and resources, so it is practically difficult. In order to operate well under a real situation, a model trained under limited operating conditions should have similar performance under other operating conditions.

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2. Literature Review

Existing research on building FDD can be divided into two main categories, including data-driven method and knowledge driven-base method [4]. The knowledge driven-based methods show a powerful capacity in simulating the diagnostic thinking of experts [4]. They showed a powerful capacity in learning patterns from training data. But they need a large amount of training data and have problems in reliability and robustness [4].

Among the classification-based methods, SVM and ANN-based methods are popular. Liang and Du developed a multi-layer SVM-based FDD method for chillers [5]. The method was effective to diagnose the faults of recirculation damper stuck, cooling coil fouling/block and supply fan speed decreasing. Dehestani et al. extended Liang and Dus work by training SVM classifiers incrementally using online monitoring data [6]. Han et al. developed a SVM-based FDD method for water-cooled centrifugal chillers [7][8][9]. The method achieved 95% correct ratio of fault diagnosis for all seven typical chiller faults on ASHRAE RP-1043 data set. Lee et al. introduced an artificial neural network to learn the relationships among the dominant symptoms and the

faults for air handling unit fault diagnosis [10]. Sun et al. employed independent component analysis to reduce the original data amount firstly, and then trained a back propagation neural network for variable refrigerant flow system FDD [11]. Similarly, Kocyigit developed a fuzzy inference system to detect sensor faults firstly. Based on the outputs of the fuzzy inference system, an artificial neural network was trained to diagnose faults in a vapor compression refrigeration system [12]. He et al. developed a hierarchical adaptive resonance theory-based neural network to detect faults in solar hot water systems [13][14]. However, they didn't consider generalization ability for various operating conditions of the FDD model.

3. Material

3.1. Experimental Data

In this study, a chiller data from experiments performed by ASHRAE Project RP-1043 was used to implement FDD model [15]. RP-1043 used a 90-ton centrifugal chiller to simulate seven different types of faults with four severity levels.

Table 1. List of tested operating conditions

Index	TEO (°C)	TCI (°C)	Capacity (%)
1	50	85	90~100
2	50	85	50~60
3	50	85	25~40
4	50	75	90~100
5	50	75	50~60
6	50	75	25~40
7	50	70	70~80
8	50	65	45~50
9	50	62	25~35
10	45	85	90~100
11	45	85	50~60
12	45	85	25~40
13	45	75	90~100
14	45	75	50~60
15	45	75	25~40
16	45	70	70~90
17	45	62	45~50
18	45	62	25~35
19	40	80	90~100
20	40	80	50~60
21	40	80	25~40
22	40	70	90~100
23	40	70	50~60
24	40	70	25~40
25	40	65	70~80
26	40	62	45~50
27	40	62	25~35

* TEO: chilled water temperature
* TCI: condenser water entering temperature
* Capacity: the chiller cooling load

The data consists of sixty-four features and recorded at 10 second time interval. The data includes both transient and steady state conditions for the following faults: loss of water flow in the condenser, loss of water flow in the evaporator, refrigerant leakage, refrigerant overcharge, presence of excess oil, condenser fouling, presence of non-condensables in the refrigerant, and faulty expansion valve. In RP-1043, the chiller was tested at 27 different operating conditions under both normal and faulty states [15]. Table 1 shows a detail of the operating conditions.

3.2. Data Preprocessing

RP-1043 has four severity level for each fault type. In this paper, the model is trained and verified using the data of the strongest fault severity level. Training the model using all features of the data can increase the dimension unnecessarily. In this paper, seven features were selected and used for model training and validation using a back-tracing sequential forward feature selection (BT-SFS) algorithm used in previous studies [16]. Information about the selected features is shown in Table 2.

Table 2. Selected features for FDD

Variable	Description
TCI	Temperature of Condenser Water In
TCO	Temperature of Condenser Water Out
TEI	Temperature of Evaporator Water In
TEO	Temperature of Evaporator Water Out
TCA	Condenser Approach Temperature
PO_feed	Pressure of Oil Feed
Evap Tons	Calculated Evaporator Cooling Rate

According to the survey of ASHRAE RP-1043, this paper selected the seven most harmful faults in terms of frequency of occurrence and economic losses [15]. The types of selected faults are as follows:

1. Condenser fouling
2. Excess oil
3. Reduced condenser water flow rate
4. Reduced evaporator water flow rate
5. Non-condensable in the refrigerant
6. Refrigerant leak/undercharge
7. Refrigerant overcharge

Transient operation is difficult and time-consuming to model and predict [17]. Therefore, steady-state detection was conducted during preprocessing. In this study, a steady-state detector from the previous study is used to filter the transient data [18][19]. Three variables, i.e., TEI, TEO and TCI, are selected as the state characteristics. If the weighted deviation of each

of these three features falls below a threshold, the chiller can be considered as under steady state. In this study, the threshold is set to 0.1°C and the time window is set to 80 s. In order to validate the generalization performance for the operating conditions of the model, only nine operating conditions were selected for training. Of the nine selected datasets, 80% were randomly selected for training and the remaining 20% were used for test. The remaining data of eighteen operating conditions were used as test data.

4. Model Establishment

4.1. Model Description

In machine learning, an artificial neural network (ANN) is a mathematical model that imitates the structure of biological neural networks [20]. An ANN consists of three main parts: structure, activation function, and learning rules [20].

The neural network consists of an input layer, an output layer, and one hidden layer. The input layer receives seven features selected in 3.2. The hidden layer is made up of ten nodes, and the output layer consists of eight nodes representing seven types of fault and one normal state. Batch normalization and dropout (regularization rate = 0.2) were applied to increase the generalization performance. To solve the gradient vanishing and increase the convergence speed, Exponential Linear Units (ELU) is used as an activation function. For the output layer, softmax which is suitable for the classification problem was used. Fig.1 depicts the ANN model structure. All models were built through the Python environment.

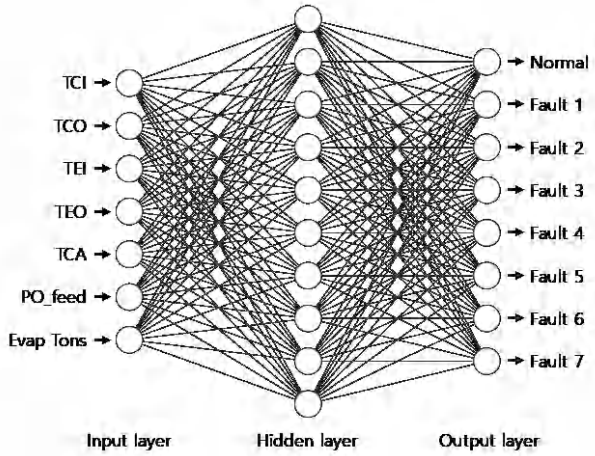


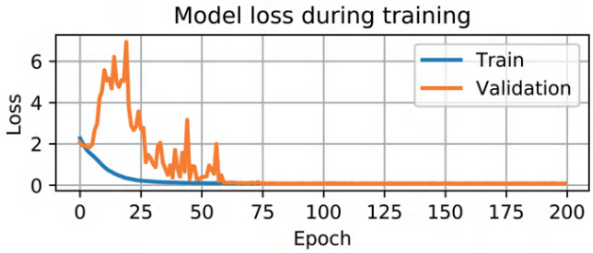
Fig.1. Structure of ANN

4.2. Model training

The model was trained using the trainset. The epoch is set to 200, and the batch size is set to 128. The RMSProp is used as the optimizer and categorical cross entropy is used as the loss function. Fig.2 shows the training process. The validation loss shows the model's performance to the validation data which was randomly selected 20% from the training set. The training process revealed that as learning progresses, the difference between validation loss and train loss decreases and finally converges closely. This demonstrates that the model did not exhibit overfitting during the training process. Also, both the validation loss and training loss progressively declined during the training, indicating that the model was appropriately trained.

The SVM-based FDD model for comparison uses the RBF kernel. The parameters of SVM was selected through a grid search. Input values of SVM were standardized during preprocessing.

Fig.2. Model loss during training



4.3. Model Validation

The purpose of this study is to propose the FDD model for considering various operating conditions. Three scenarios were introduced to validate the performance of the model under various operating conditions. The detail for each scenario is shown in Table 3. In scenario 1, the model was trained using the data when the TCI is set to 50°C and then tested using the data at the remaining operating conditions. Scenario 2 and 3 introduced each in the same way with 45°C and 40 °C of TCI.

Table 3. Description of scenarios

Scenario	Operating conditions for training
1	1~9
2	10~18
3	19~27

Fig.3, 4, and 5 show the test results for scenarios 1, 2, and 3, respectively. The yellow background section shows an accuracy of the test set under the operating conditions used for the training and the other section shows the accuracy of the test set under the operating conditions used for the testing. The red bar shows the classification accuracy of the SVM-based model, and the blue bar shows the classification accuracy of the ANN-based model. Fig.3, 4 and 5 show that at the trained operating conditions, both models show highaccuracy. However, under untrained operating conditions, it shows that the accuracy of the ANN-based model is higher than the SVM-based model. Table 4 shows the average classification accuracy of each model. Compared to the SVM-based model, accuracy of the ANN-based model is higher by about 81%, 110%, and 76% for scenarios 1, 2, and 3, respectively

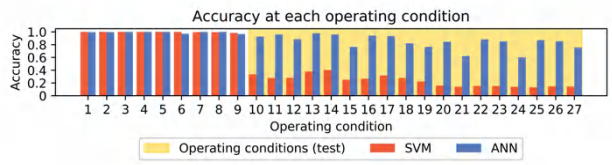


Fig.3. Classification result (scenario 1)

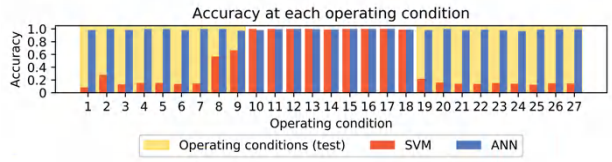


Fig.4. Classification result (scenario 2)

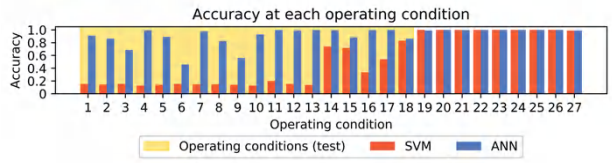


Fig.5. Classification result (scenario 3)

Table 4. Average accuracy of each model

Scenario	Accuracy of SVM-based model	Accuracy of ANN-based model
1	0.49	0.89
2	0.47	0.99
3	0.52	0.92

5. Conclusions

ANN-based model training requires appropriate dataset of normal and fault situation. It is difficult to obtain eligible data under all operating conditions. For practical use, a model trained under limited operating conditions should have acceptable performance under untrained operating conditions. In this study, the ANN-based chiller FDD model for considering various operating conditions was proposed. The proposed model generalizes well under various operating conditions as it can find underlying patterns in the data. Experiments with the data from ASHRAE 1043-RP show that the proposed model has improved generalization performance for operating conditions compared to the SVM-based model. However, experiment was conducted using only one dataset, and considering that the data was produced in the laboratory, it is necessary to do a case study in the actual building in future work.

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